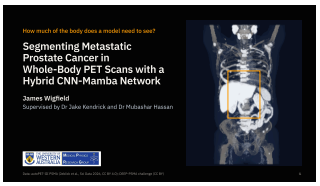
How Much of the Body Does a Model Need to See?: script

Segmenting Metastatic Prostate Cancer in Whole-Body PET Scans with a Hybrid CNN-Mamba Network

12 main slides · 1791 words · 13:43 spoken at 135 words a minute (including 27 s of clips without narration) · target 13:50



`Slide 1 · 22 s · ends at 0:22 · 51 words`

1. Opens on the title. Nothing to click.

Hi everyone, I'm James. My project uses a hybrid network, made of convolutions and a newer layer called Mamba, to find prostate cancer in whole-body PET scans. My supervisors are Jake Kendrick and Mubashar Hassan. My talk asks one question: how much of the body does a model need to see?

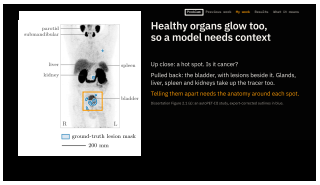


`Slide 2 · 98 s · ends at 2:00 · 201 words`

Outlining every lesion by hand is slow, and experts disagree

1. Opens on case A, facing front. Ask the room to look.
2. Click: the five-second countdown runs round the scan. Stay quiet.
3. Click: the ground-truth clip of case A plays (the reveal). Keep talking over it.
4. Click: the cost of outlining by hand and the two-observer Dice.
5. Click: the talk in one breath, beside the plain scan.

Here's a whole-body PSMA PET/CT scan. PSMA is a protein that prostate cancer cells carry in large amounts. The PET shows where a radioactive tracer that binds to it has collected, and the CT shows the anatomy. When the cancer spreads, it forms lesions: small spots of disease scattered through the body. `CLICK` You've got five seconds. See how many you can spot. `PAUSE WHILE THE COUNTDOWN RUNS` `CLICK` `CLIP PLAYS, 8.8 S, KEEP TALKING` There are 52, each marked in the reference labels, which were drawn from a threshold and corrected by an expert. The blue is my model's output; more on that later. `CLICK` Outlining every lesion like this is called segmentation, and by hand it's slow. Experts also disagree. When a second observer re-outlined 28 scans, the two sets of outlines overlapped by only 33% on average. That overlap score is called Dice. `CLICK` Here's the talk in one breath. Healthy organs glow too, so context matters. Mamba might let a network see more of it. My network adds Mamba to its three deepest stages, with an organ task beside it. On 697 scans, with 130 held out, it found 82% of lesions, which is its sensitivity. And 86% of what it flagged was real, which is its precision. Then comes a twist. `(cut if behind)`

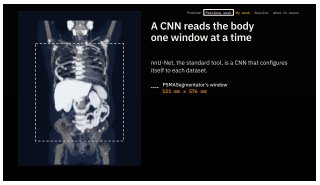


`Slide 3 · 46 s · ends at 2:46 · 100 words`

Healthy organs glow too, so a model needs context

1. Opens zoomed in on the bladder region of panel (a). Ask: cancer?
2. Click: the camera pulls back to the whole body and its organ labels.
3. Click: the context line and the autoPET-III citation.

Up close, this is just a hot spot. Is it cancer? `CLICK` Pull back, and it's the bladder, with lesions sitting right next to it. PSMA tracers also collect in healthy organs: the salivary glands, liver, spleen and kidneys, and the bladder, where the tracer drains out. Urine in the bladder can even hide small lesions near the prostate. `(cut if behind)` So brightness alone doesn't make a lesion. You need to know where you are in the body. `CLICK` A winning autoPET-III team made the same argument: telling disease from normal uptake needs the surrounding anatomy. That's the problem my organ task tackles later.

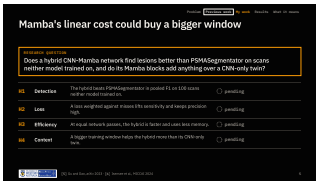


`Slide 4 · 76 s · ends at 4:02 · 168 words`

A CNN reads the body one window at a time

1. Opens zoomed in on the window, then pulls back to the body.
2. Click: GPU memory caps the window.
3. Click: the 2022 cascade figures, as context only.

The standard tool is nnU-Net. It's a convolutional neural network, or CNN, that configures itself to each new dataset. A convolution looks at a small patch of nearby voxels, which are 3D pixels, and its view widens layer by layer. A whole-body scan is too big to process at once, so the network reads it one 3D window at a time. The dashed box is the window of PSMASegmentator, a later nnU-Net model, drawn to scale on this patient. `CLICK` A bigger window gives more context, but GPU memory, the memory on the graphics card, caps it. A bigger window means a smaller batch, which is how many windows it trains on at once, or a shallower network. `CLICK` The 2022 nnU-Net cascade from Kendrick and colleagues is where this project started. On its own 128 test scans, its precision was 88.2% and it found 73.0% of lesions. That's a different test set, so I only use those figures as context. Its authors named sensitivity as the thing to improve. `(cut if behind)`

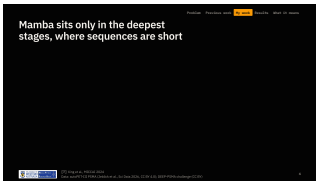


`Slide 5 · 68 s · ends at 5:10 · 154 words`

Mamba's linear cost could buy a bigger window

1. Opens on the attention-against-Mamba schematic.
2. Click: the U-Mamba warning and the CNN-only twin.
3. Click: the research question, large. Pause on it.
4. Click: the four hypotheses, verdicts pending.

Mamba is a newer kind of sequence model. Attention, the core of a transformer, compares every position in a sequence with every other one. So its cost grows with the square of the length. Mamba makes one pass with a small running memory, so its cost grows linearly. My proposal's bet was that this would let a hybrid train on bigger windows than nnU-Net, and so see more context. `CLICK` When another group switched off the Mamba layers in U-Mamba, the results didn't change. So any claim for Mamba needs a twin: the same network with the Mamba blocks taken out. `CLICK` That gives my research question. `PAUSE` It has two parts. Is the hybrid better than PSMASegmentator on scans neither has seen? And if it is, is the reason Mamba, or would the twin do just as well? `CLICK` I tested four hypotheses, on detection, the loss, speed and context. I'll give the verdicts at the end. `(cut if behind)`

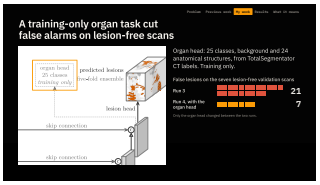


`Slide 6 · 62 s · ends at 6:12 · 139 words`

Mamba sits only in the deepest stages, where sequences are short

1. Opens on case A with my window and the dashed PSMASegmentator window.
2. Click: the camera moves into the window and onto the network.
3. Click: zoom to the three deepest stages (blue).
4. Click: the six token lanes with their real lengths.

Here's my window on the same patient. It's 20 cm wide and 29 cm tall, smaller than PSMASegmentator's, because it was the largest that fitted my development GPU. `CLICK` Inside the window is the network. It's a U-shaped encoder and decoder with five stages, and every stage has convolutions. `CLICK` The three deepest stages, in blue, also have Mamba blocks. Each stage halves the grid, so down here the feature map is small. `CLICK` A Mamba block flattens the feature map into a line of tokens, one per voxel. It scans that line in six directions, forwards and backwards along each axis. That's 27,648 tokens at the third stage, down to 432 at the fifth. At the second stage it would be 221,184, too costly to scan. `(cut if behind)` So Mamba adds context inside the window. It never sees the whole body at once.

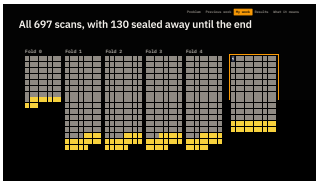


`Slide 7 · 46 s · ends at 6:58 · 102 words`

A training-only organ task cut false alarms on lesion-free scans

1. Opens on the organ head of the network (training only).
2. Click: false lesions on the lesion-free validation scans, run 3 against run 4.

There's one more piece. Beside the lesion output, a second head labels 25 classes: background and 24 anatomical structures, taken from automatic CT segmentations. A network that knows where the kidneys and bladder are should raise fewer false alarms there. PSMASegmentator's recipe uses the same idea. Mine is only used in training and is switched off at inference. `CLICK` I added it after my third training run, run 3, flagged lesions on all seven lesion-free scans in my validation set. With the organ head, the false lesions on those scans fell from 21 to 7. Everything else in that run stayed the same. `(cut if behind)`

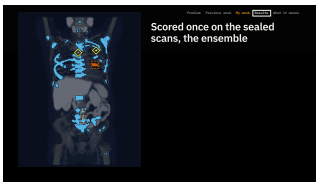


`Slide 8 · 60 s · ends at 7:58 · 133 words`

All 697 scans, with 130 sealed away until the end

1. Opens on one tile, then pulls back to every scan.
2. Click: the sealed test scans slide into the vault.
3. Click: the pool splits into five folds, cohorts kept.

Each tile is one scan, 697 in all, from two public datasets. autoPET-III's outlines were corrected by an expert. DEEP-PSMA's mark every voxel at or above a fixed uptake threshold. Neither dataset is limited to one tracer or to recurrent disease. `(cut if behind)` `CLICK` I sealed 130 scans away as the test set. Nothing trained on them, and they were scored once. `CLICK` The other 567 went into five-fold cross-validation. Each of five models trains with a different fifth held out for checking. Averaged together, the five form an ensemble, which I call run 5. Before the test, I wrote down the cut-off for calling a voxel a lesion, the smallest blob I'd keep, and what I'd report. That's what pre-registered means here. Run 4, which comes up later, is one earlier hybrid trained on autoPET-III alone.



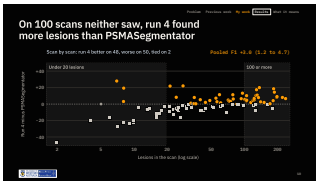


`Slide 9 · 56 s · ends at 8:54 · 122 words`

Scored once on the sealed scans, the ensemble found most lesions

1. Opens on case A with its verdict counts.
2. Click: the pooled score over every sealed scan.

Back to the patient from the start, one of those sealed scans. `(cut if behind)` The model found 49 of its 52 lesions, missed 3, and raised 2 false alarms. `CLICK` Over all 130 sealed scans, the ensemble's F1 was 84.3%. Sensitivity, the share of lesions found, was 82.4%. Precision, the share of what it flagged that was real, was 86.2%. F1 balances the two. A lesion counts as found if any predicted voxel touches it. That rule is generous, so I report Dice beside it, here 59.6%. Two cautions. Lesions are pooled, so scans with lots of disease count the most. And it's in-domain, because both datasets were in training. PSMASegmentator trained on all 110 autoPET-III test scans, so I can't fairly score it here.

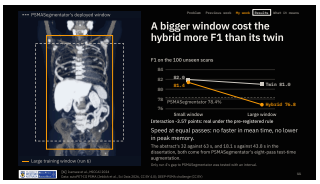


`Slide 10 · 76 s · ends at 10:10 · 166 words`

On 100 scans neither saw, run 4 found more lesions than PSMASegmentator

1. Opens on the DEEP-PSMA scans and the underdog card.
2. Click: run 4 minus PSMASegmentator, pooled.
3. Click: the tiles become one dot per scan.

For a fair test I needed scans neither model had seen. These are the 100 DEEP-PSMA scans. PSMASegmentator never trained on them, and neither did run 4. Run 4 is one model trained on 430 scans. PSMASegmentator averages five models trained on 1,035. Each runs eight mirrored passes, predicting the scan again flipped and averaging. `CLICK` Run 4 still scored 3.0 F1 points higher, and the interval, 1.2 to 4.7, stays above zero. Its sensitivity was 8.1 points higher, but its precision was 6.4 points lower. So it's a trade: more lesions found, for more false alarms. `CLICK` Scan by scan, there was no clear difference. Run 4 was better on 48 scans and worse on 50. It lost on 18 of the 22 scans with under 20 lesions, and won on 23 of the 28 with 100 or more. `(cut if behind)` The pooled gain comes from the scans with the most disease. `PAUSE` This was a fair test of two recipes, and most of the differences between them favour PSMASegmentator.



`Slide 11 · 98 s · ends at 11:48 · 223 words`

A bigger window cost the hybrid more F1 than its twin

1. Opens on the small window and the two small-window scores.
2. Click: the window grows to the large patch while the lines fall.
3. Click: the PSMASegmentator line and the speed lines.

My main models used the small window, so they never tested the bet. Run 6 did, with an ablation: take one part out, here the Mamba blocks, and see what changes. That leaves a CNN-only twin. It has 42% of the hybrid's parameters, but that gap is the same at both window sizes. `(cut if behind)` At the small window, on the same 100 unseen scans, the hybrid scored 81.4 and the twin 82.0. `CLICK` Then I grew the window to 4.7 times the volume, longer head to toe than PSMASegmentator's. The hybrid was predicted to gain more. Instead, it fell to 76.8, while the twin barely moved, to 81.0. The hybrid lost 3.57 points more than the twin. Under the rule I fixed before scoring, that's a real effect. One possible reason is length: at the big window the third stage scans 129,024 tokens through the same small memory. I didn't test why. `CLICK` The twin also scored above PSMASegmentator's 78.4%, so I can't credit run 4's gain to Mamba. Only run 4's gap was tested with a confidence interval, though. On speed, the abstract's 32 against 63 seconds, and the 18.1 s against 43.8 s in my dissertation, both come from PSMASegmentator's eight-pass test-time augmentation. In my dissertation's timing, at equal passes, the hybrid was no faster in mean time and no lower in peak memory.

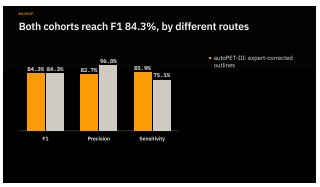


`Slide 12 · 122 s · ends at 13:50 · 232 words`

As trained, a bigger window helped neither; the gain is not Mamba's

1. Opens on the answer and the four verdicts with their evidence.
2. Click: limitations, with case B.
3. Click: the next step; say the demo line, then let the recording run.
4. Click: thank you and questions; leave this up for questions.

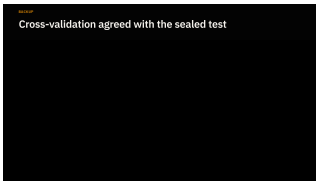
So how much of the body does a model need to see? On my evidence, a bigger window, as trained, helped neither model, and it hurt the hybrid most. Adding context this way didn't pay off. And run 4's gain over PSMASegmentator isn't credited to Mamba, because the CNN-only twin also scored above it. On the hypotheses, detection held, but only as a trade. The loss didn't hold: weighting misses found 71% of lesions on validation but cut precision to 61%. Speed wasn't supported either. And context came out reversed on DEEP-PSMA, my primary set. The project offers a fair test on unseen scans, and a pre-registered test of Mamba. `CLICK` This is case B, the lowest F1 by a fixed rule. Both of its lesions were found, but it also raised 14 false alarms. Of those, 8 were in bone. `(cut if behind)` Each run 6 cell is a single training run, and the big window trained at batch size 1. The two datasets label lesions differently, and only 8 test scans were lesion-free. None of this is clinical validation. `CLICK` The next step is the Sir Charles Gairdner Hospital cohort my proposal named. The code will be released, so the UWA Medical Physics Research Group can retrain the model and score it there. Here's a real run of the five models in the app I built, sped up. `CLIP PLAYS, 18 S, LET IT RUN` `CLICK` Thank you. I'm happy to take questions.



`Backup B1 · 76 words`

Both cohorts reach F1 84.3%, by different routes

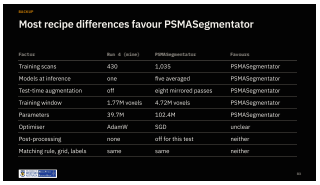
The two cohorts label lesions differently, so I report them separately. autoPET-III outlines were corrected slice by slice by an expert. DEEP-PSMA marks every voxel at SUV 3 or more, with normal uptake painted out, so a scan splits into about 68 small pieces. The likely reason for the gap: on DEEP-PSMA nearly everything the model predicts lands on disease, so precision is 96.0%, but it misses many small pieces. Both land on the same F1.



`Backup B2 · 63 words`

Cross-validation agreed with the sealed test

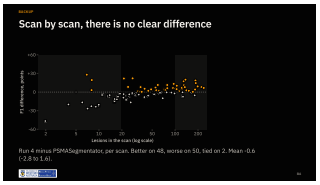
Before the test, each of the 567 pool scans was predicted by the fold model that never trained on it. That pooled F1 was 81.8%, with a fold-to-fold spread of 1.5 points, and it sits inside the test interval of 81.4 to 86.5. A single fold model scored 83.1 on average on the test set, so averaging the five added about 1.2 points.



Backup B3 · 70 words

Most recipe differences favour PSMASegmentator

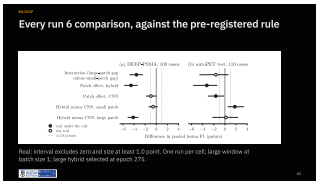
This is the honest list of what differs. PSMASegmentator trained on 1,035 scans against my 430, averages five models with eight mirrored passes, uses a bigger window, and has 102.4M parameters against 39.7M. It also has deep supervision and a misalignment augmentation that I did not port. Most factors favour it, so my gain is more likely to understate than overstate the recipe, but none was tested on run 4.



Backup B4 · 65 words

Scan by scan, there is no clear difference

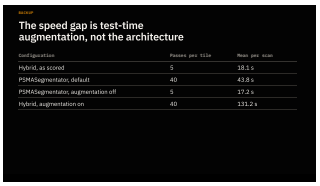
Each dot is one of the 100 DEEP-PSMA scans. Pooled F1 weighs lesion-heavy scans most, so the pooled gain comes from the right of this plot. Run 4 was below PSMASegmentator on 18 of the 22 scans with under 20 lesions and above it on 23 of the 28 with 100 or more. The mean per-scan difference was -0.6 points, with an interval spanning zero.



Backup B5 · 62 words

Every run 6 comparison, against the pre-registered rule

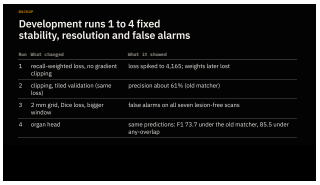
These are all five comparisons I committed to before scoring, from the dissertation. On DEEP-PSMA the interaction was -3.57 points, from -4.72 to -2.48, a real effect under the rule. On autoPET it was -1.62, no clear effect. The caveats: one training run per cell, the large window trained at batch size 1, and the large hybrid was selected at epoch 275.



Backup B6 · 73 words

The speed gap is test-time augmentation, not the architecture

Timed on one RTX 4090 over 20 test scans, each model already loaded. As scored, the hybrid took 18.1 s against 43.8 s for PSMASegmentator, but that gap is its eight-pass augmentation. At equal passes the hybrid was no faster in mean time and used 5.2 GiB of memory against 4.5 GiB. The abstract quoted 32 against 63 seconds from an earlier run on my home RTX 5070 Ti; the same caveat applies.



Backup B7 · 76 words

Development runs 1 to 4 fixed stability, resolution and false alarms

Run 1 had no gradient clipping and lost about three of its twelve hours to loss spikes. Run 2 added clipping, but the recall-weighted loss left precision at about 61%. Run 3 moved to the finer grid and Dice, then flagged every lesion-free validation scan. Run 4 added the organ head. Runs 2 to 4 were scored under an older one-to-one matcher; on identical run 4 predictions it gave 73.7 where the adopted rule gives 85.5.



Backup B8 · 64 words

Mamba sits only where the flattened sequence is short

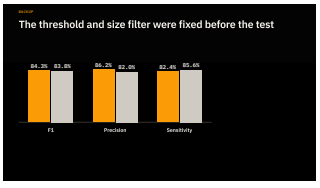
Each deeper stage halves the grid, so the sequence a Mamba layer would scan shrinks fast. The shallow stages would be at least 221,184 tokens per direction, and their features are fine texture a convolution handles cheaply. In the deep three, each block scans six directions and sums them. I never tested the alternatives, such as alternating directions across layers or a larger state.



Backup B9 · 68 words

Case B: both lesions found, 14 false alarms

Case B was chosen by rule before I looked: the lowest F1 among scans where F1 is defined. It is not the worst scan, because nine scans had no lesion found and so have no F1. Both of its lesions were found. Its 14 false alarms were 8 in bone and 6 in soft tissue, and none sat in the kidneys or bladder, which the organ task targets.



Backup B10 · 64 words

The threshold and size filter were fixed before the test

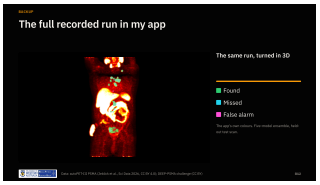
The operating point came from the five-fold validation sweep, before any test inference. With the 10-voxel filter, F1 moved by less than a point across thresholds, so I kept 0.5. The protocol fixed two rows: the filtered headline and an unfiltered row, because PSMASegmentator has no filter. On the test set the filter trades a little sensitivity for precision, and both rows are reported.



Backup B11 · 27 words

References

These are the works cited on the slides, in the order they appear, in the same IEEE style as the dissertation. The dissertation has the full list.



`Backup B12 · 62 words`

The full recorded run in my app

This is a real run of the five-model ensemble in the app I built, on a held-out test scan, sped up 6x. It needs a CUDA GPU, so I recorded it on my PC. The colours are the app's own: found, missed and false alarm. The code will be released so the UWA Medical Physics Research Group can retrain and score it.